R - 27/10/26

To prove

~ Bayesian NE ⇒ Nash eq (always exists)

Agent form game

Γ : $\langle N, (T_i)_{i \in N}, p, S, (A_t)_{t \in T_i} \rangle$ is an incomplete information game.

We formulate an agent form game $\hat{\Gamma}$

$\hat{\Gamma} = \langle \hat{N}, (S_t)_{t \in \hat{N}}, \hat{u}_t \rangle$

- $\hat{N} = \bigcup_{i=1}^{N} T_i$
- $\hat{S}_t = A_i(t_i)$ where $t = t_i$
- Assume $t = t_i$ $t_{-i} | t_i$

  then $\hat{u}_t(a) = \sum_{t_{-i} \in T_{-i}} f(t_{-i}|t_i)\, u_i\left((t_i, t_{-i}), (a_j(t_j))_{j \in N}\right)$

Remark: For a player $i \in N$, a behaviour strategy σ_i gives a mixed strategies of $t \in T_i$. Also every mixed strategy of the players in T_i will give a behaviour strategy for player i in the incomplete information game.

Theorem: A strategy vector $(\sigma_i)_{i \in N}$ in the incomplete information game is a Bayesian NE iff the strategy vector $(\sigma_i(t_i))_{i \in N,\, t_i \in T_i}$ is a NE in the corresponding agent form game.

Idea of proof: Suppose $(\sigma_i)_{i \in N}$ is not a BNE

$\exists\, i \in N$ and $t_i \in T$ s.t.

$u_i(\sigma | t_i) < U_i((a, \sigma_{-i}) | t_i)$ for some $a \in A_i(t_i)$

We can show that $(\sigma_i(t_i))_{i \in N,\, t_i \in T_i}$ is not a NE as player t_i will

Deviate to action a

$$\hat{U}_{t_i}\left((\sigma_i(t_i))_{i\in N, t_i\in T_i}\right) < \hat{U}_{t_i}\left((a, \sigma_{-i})\right)$$

If the agent form game is a finite game, it has an NE in mixed strategies and that NE is a BNE in the incomplete information game.

Ex: $N_1 = \{1, 2\}$

$T_1 = \{t_1, t_2\}$, $T_2 = \{\text{II}\}$

$p(t_1, \text{II}) = 1/2$, $p(t_2, \text{II}) = 1/2$

	L	R
T_1	(1,0)	(0,2)
B_1	(0,3)	(1,0)

(t_1, II)

	L	R
T_2	(0,2)	(1,1)
B_2	(1,0)	(0,2)

(t_2, II)

$\hat{N} = \{t_1, t_2, \text{II}\}$

$S_{t_1} = \{T_1, B_1\}$

$S_{t_2} = \{T_2, B_2\}$

$S_{\text{II}} = \{L, R\}$

$\hat{U}_{t_1}(T_1, B_2, L) = 1 \times u_1\left((t_1, \text{II}), (T_1, L)\right) = 1$

$\hat{U}_{\text{II}}(T_1, B_2, L) = u_2\left((t_1, \text{II}), (T_1, L)\right) p(t_1|\text{II}) + u_2\left((t_2, \text{II}), (B_2, L)\right) p(t_2|\text{II})$

$= 0 \times p(t_1|\text{II}) + p(t_2|\text{II}) \times 0$

$= 0$

$p(t_1|\text{II}) = \frac{p(t_1, \text{II})}{p(\text{II})} = 1/2$

q $1-q$

L R

$\sigma = [x(T_1), (1-x)(B_1)] , [y(T_2), (1-y)(B_2)]$

Suppose this strategy profile is not a BNE

Suppose player 1 has a profitable deviation at t_1, say he can play T_1 & get a better utility.

$$U_1(\sigma) = p(t_1)\, u_1(\sigma|t_1) + p(t_2)\, u_1(\sigma|t_2)$$

$$U_1(\sigma|t_1) < U_1((T_1, \sigma_2)|t_1)$$

$$\bar{\sigma}_1(t_1) = \begin{cases} T_1 & \text{if } t_1 = t_1 \\ [y(T_1), (1-y)(B_2)] & \text{if } t_1 = t_2 \end{cases}$$

$$U_1(\sigma) < U_1(\bar{\sigma}_1(t_1), \sigma_2)$$

Theorem: In Harsanyi incomplete info. game, every NE is also a Bayesian NE.

Correlated Equilibrium

2 can C → Cross S → stop.

	C	S
C	-10, -10	5, 0
S	0, 5	-1, -1

PS NE (C, S), (S, C)

MS NE $\left(\frac{3}{8}[C], \frac{5}{8}[S] \;;\; \frac{3}{8}[C], \frac{5}{8}[S]\right)$

$\left(\frac{-5}{8}, \frac{-5}{8}\right)$

	C	S
C	0	0.55
S	0.4	0.05

} P.D of signal.

Players know the prob. distribution.

For P-1 suppose the signal shows C

If P1 obeys the signal and P-2 ~~shows~~ also does the same, P-1 utility will be 5.

If P1 does not, the utility will be -1.

For P1 the signal shows S

$$p(C|S) = \frac{p(C,S)}{p(S)} = \frac{0.4}{0.45} = 0.89 \qquad p(S|S) = \frac{0.05}{0.45} = 0.11$$

If P1 follows the signal and stops, $0.89 \times 0 + 0.11 \times -1 = -0.11$.

Satya, Amit

2e base

If player 1 does not follow & play crosses

$0.89 \times -10 + 0.11 \times 5 = -8.9 + 0.55 = -8.35$

P-1

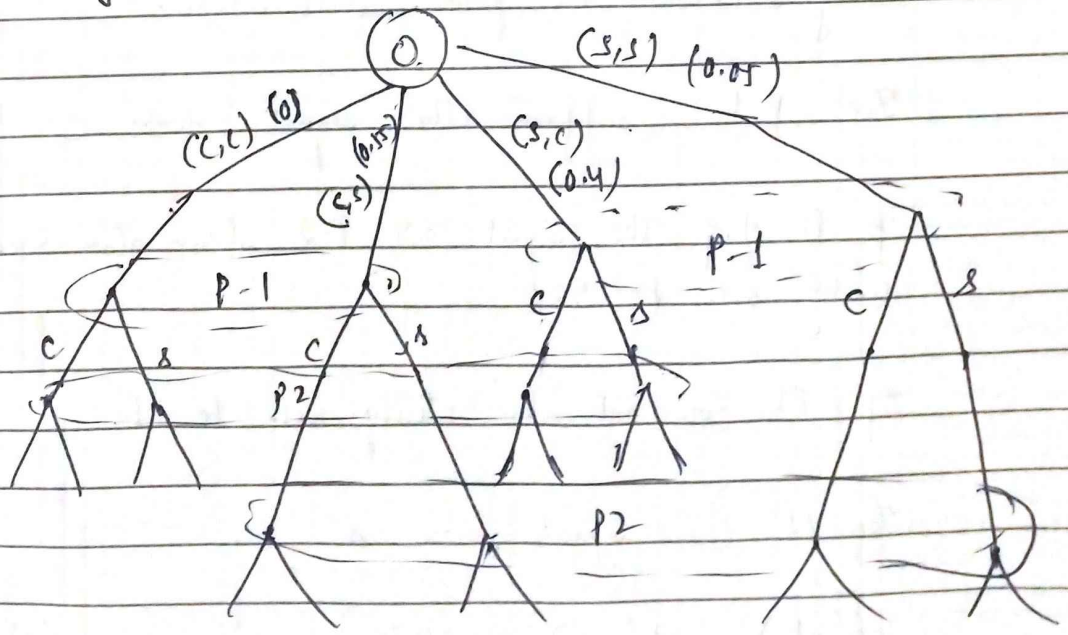
$5 \times 0.55 + 0.4 \times 0 + -1 \times 0.05 = 2.70 > \frac{-5}{8}$

$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ Let $S = \prod_{i=1}^{n} S_i$ and p be a prob distribution on S

An outcome observer probabilistically chooses a strategy profile $s = (s_1, s_2, \ldots, s_n)$ according to p.

To each player he reveals s_i, but not s_{-i}. This is the recommendation of the observer for player i.

- Each player i chooses an action $s_i' \in S_i$ (possibly different from s_i). The player i recieves the utility $u_i(s_1', s_2', \ldots, s_n')$.



The extensive form game corresponding to p is denoted by $\Gamma^*(p)$

In player i, ~~the strategies are~~

a pure strategy $\tau_i : S_i \to S_i$

the strategy $\bar{\tau}_i(s_i) = s_i \quad \forall s_i \in S_i$ is the strategy corresponding to following recommender's action

The prob dist. p is a correlated eqbm in a game G if $(\bar{\tau}_i)_{i \in N}$ is a NE in the game $\Gamma^*(p)$.

The recommendations are private and this is imp. for the def.

Theorem:-

The strategy vector $\bar{\tau}$ is a NE of the game $\Gamma^*(p)$ iff

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(\bar{\tau}_i(s_i), s_{-i}) \geq \sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})\, u_i(s_i', s_{-i})$$

$\forall i \quad \forall s_i \quad \forall s_i'$

Proof:- ~~Suppose player i recieves recommendation s_i~~

Fix i, s_i, s_i'

Case-1

$$\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) = 0 \longrightarrow p(s_i, s_{-i}) = 0 \quad \forall s_{-i}$$

$$\Rightarrow p(s_i) = 0$$

Case-2 $\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i}) > 0$

If Player i recieves s_i

$$p(t_{-i} \mid s_i) = \frac{p(s_i, t_{-i})}{\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})}$$

The utility of player i by following the recommendation s_i & assuming others are also following the recommendation.

$$\sum_{t_{-i} \in S_{-i}} p(t_{-i} \mid s_i) u_i(s_i, t_{-i})$$

If player deviates to s_i' believing others are following the recommendation

$$\sum_{t_{-i} \in S_{-i}} p(t_{-i} \mid s_i) u_i(s_i', t_{-i})$$

$\tilde{\tau}$ will be a NE iff $\sum_{t_{-i} \in S_{-i}} p(t_{-i} \mid s_i) u_i(s_i, t_{-i})$.

$$\geq \sum_{t_{-i} \in S_{-i}} p(t_{-i} \mid s_i) u_i(s_i', t_i)$$

$\forall i \quad \forall s_i \quad \forall s_i'$

$$\sum_{t_i \in S_i} \sum_{t_{-i} \in S_{-i}} \frac{p(s_i, t_{-i})}{\sum_{s_{-i} \in S_{-i}} p(s_i, s_{-i})} u_i(s_i, t_i) \geq \sum_{t_i \in S_{-i}} \frac{p(s_i, t_{-i})}{\sum_{s_{-i}} p(s_i, s_{-i})} u_i(s_i', t_i)$$

Bayes

$$\Rightarrow \sum_{t_{-i} \in S_{-i}} p(s_i, t_{-i}) u_i(s_i, t_{-i}) \geq \sum_{t_{-i} \in S_{-i}} p(s_i, t_{-i}) u_i(s_i', t_i)$$

In the given ex

0	1
0	0

is a correlated eqb^m

0	1-α
α	0

is also a corr eqb^m.

Theorem: Every mixed strategy NE induces a correlated eqb^m.

σ^* is an NE

$$p_{\sigma^*}(s_1, s_2, \ldots, s_n) = \sigma_1^*(s_1), \sigma_2^*(s_2) \ldots \sigma_n^*(s_n)$$

p_{σ^*} is a correlated eqb^m.

Prisoner's Dilemma

	C	D
C	(3,3)	(0,5)
D	(5,0)	(1,1)

(D,D) is the expected outcome

Is there a correlated eqb^m where the strategy profile (C,C) gets a positive probability.

Satya
Amit
Subhra.

AUCTION THEORY

11

For a real valued RV X, we define the CDF of X, F_X, is defined by

$$F_X(c) = \mathbb{P}(X \leq c) \quad , \forall c \in \mathbb{R}.$$

If X is absolutely ~~continuity~~ cont. RV, then there exists a pdf of X. f_X is a f^n, $f_X : \mathbb{R} \to \mathbb{R}^+$ s.t $\forall c \in \mathbb{R}$.

$$F_X(c) = \int_{-\infty}^{c} f_X(c)\, dx.$$

- **Sealed-bid auction with independent private valuations**

Def^n A sealed-bid auction with independent private values is a vector $(N, (V_i, F_i)_{i \in N}, p, c)$

where

$N = \{1, 2, \ldots, n\}$ set of buyers

- V_i = The set of all valuations for player i where $V_i \in \mathbb{R}$.
- F_i = For player i, F_i denotes the CDF of player i's valuation for the object.
- $p : [0, \infty)^N \to \Delta N$ denotes the probability of players winning the object given the profile of bids.

  ΔN is the set of prob. distributions over N.
- $c : N \times [0, \infty)^N \to \mathbb{R}^N$ is a f^n determining the payment for each buyer.

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A sealed bid auction is conducted as follows:

- The players realize their private values v_i, ℓ an element in V_i, selected according to F_i.
- Every buyer learns about their own valuation but not about others valuation.
- Every buyer submits a bid $b_i \in [0, \infty)$ in a sealed/closed envelope.
- A winner is selected according to p.
- Buyers pay the payment based on C.